

SKILL-6

6. To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

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rishitha@Rishitha: ~/skill6
rishitha@Rishitha:~$ mkdir -p skill6
rishitha@Rishitha:~$ cd skill6
rishitha@Rishitha:~/skill6$ pwd
/home/rishitha/skill6
rishitha@Rishitha:~/skill6$ nano escape.c
rishitha@Rishitha:~/skill6$ gcc escape.c -o escape
rishitha@Rishitha:~/skill6$ ./escape
Skill-6 Escape Sequence Parser
Enter input: hello\ world
Original : hello\ world
Parsed   : hello world
rishitha@Rishitha:~/skill6$ ./escape
Skill-6 Escape Sequence Parser
Enter input: hello\&world
Original : hello\&world
Parsed   : hello&world
rishitha@Rishitha:~/skill6$ ./escape
Skill-6 Escape Sequence Parser
Enter input: hello\ world\ \&\ test
Original : hello\ world\ \&\ test
Parsed   : hello world & test
rishitha@Rishitha:~/skill6$ ./escape
Skill-6 Escape Sequence Parser
Enter input: hello\\world
Original : hello\\world
Parsed   : hello\world
rishitha@Rishitha:~/skill6$ ./escape
Skill-6 Escape Sequence Parser
Enter input: hello\ world\ \&\ test\ \$100
Original : hello\ world\ \&\ test\ \$100
Parsed   : hello world & test $100
rishitha@Rishitha:~/skill6$ nano process.c
rishitha@Rishitha:~/skill6$ gcc process.c -o process
rishitha@Rishitha:~/skill6$ ./process ls
Parent Process
Parent PID: 686
Waiting for child process...
Child Process
Child PID: 687
Executing command: ls
escape escape.c process process.c
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Child exited with status: 0
Parent process completed.
rishitha@Rishitha:~/skill6$ ./process echo Hello
Parent Process
Child Process
Parent PID: 697
Child PID: 698
Waiting for child process...
Executing command: echo
Hello
Child exited with status: 0
Parent process completed.
rishitha@Rishitha:~/skill6$ ./process pwd
Parent Process
Child Process
Parent PID: 708
Waiting for child process...
Child PID: 709
Executing command: pwd
/home/rishitha/skill6
Child exited with status: 0
Parent process completed.
rishitha@Rishitha:~/skill6$ echo Hello
Hello
rishitha@Rishitha:~/skill6$ ./process pwd
Parent Process
Parent PID: 728
Child Process
Waiting for child process...
Child PID: 729
Executing command: pwd
/home/rishitha/skill6
Child exited with status: 0
Parent process completed.
rishitha@Rishitha:~/skill6$ ./process abcdxyz
Parent Process
Parent PID: 741
Waiting for child process...
Child Process
Child PID: 742
Executing command: abcdxyz
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/home/rishitha/skill6
Child exited with status: 0
Parent process completed.
rishitha@Rishitha:~/skill6$ ./process abcdxyz
Parent Process
Parent PID: 741
Waiting for child process...
Child Process
Child PID: 742
Executing command: abcdxyz
execvp failed: No such file or directory
Child exited with status: 1
Parent process completed.
rishitha@Rishitha:~/skill6$ ./process date
Parent Process
Child Process
Child PID: 790
Parent PID: 789
Executing command: date
Waiting for child process...
Sun Sep 27 10:27:32 UTC 2026
Child exited with status: 0
Parent process completed.
rishitha@Rishitha:~/skill6$ ls
escape  escape.c  process  process.c
rishitha@Rishitha:~/skill6$ ./escape
Skill-6 Escape Sequence Parser
Enter input: hello\ world
Original : hello\ world
Parsed   : hello world
rishitha@Rishitha:~/skill6$ ./process ls
Parent Process
Child Process
Parent PID: 824
Waiting for child process...
Child PID: 825
Executing command: ls
escape  escape.c  process  process.c
Child exited with status: 0
Parent process completed.
rishitha@Rishitha:~/skill6$
```